

Adaptive Reasoning and Collaboration Network

INFORMATION PAPER

SUBJECT: ADAPTIVE REASONING AND COLLABORATION NETWORK DASHBOARD

Encl: (1) Mesh and Hybrid Agent Selection
(2) Commander's Input Flow Diagram

BACKGROUND. Commanders at all levels face increasingly complex, time sensitive decisions involving combat readiness, training schedules, maintenance backlogs, and funding constraints. Existing systems provide fragmented visibility and require substantial staff effort to fuse data into coherent, auditable, and feasible courses of action (COAs). These processes are manual and time consuming; going back to the drawing board costs commanders both time and money when both are scarce.

PURPOSE. To propose the phased development of an Adaptive Reasoning and Collaboration (ARCnet) network to give commanders a unified dashboard that reduces decision latency and improves decision outcomes.

DISCUSSION.

(a) System Summary.

(1) ARCnet is a policy-bound, reasoning architecture that operationalizes agentic workflows within doctrinal structures and accountability constraints. It reflects design principles from large-scale distributed AI systems: embedding-based expert routing, intelligently gated reasoning pipelines, Retrieval Augmented Generation, and formalized human-in-the-loop (HITL) governance.

(2) The system ingests a commander's free-text request through a scribe agent that transforms it into a structured format: Mission Context/intent, end state, constraints, CCIRs, and acceptance criteria. This context serves as a shared semantic substrate for all downstream reasoning. ARCnet then facilitates three computational pathways:

a. Organizational mode which mirrors traditional staff-lane workflows and emphasizes typical G-Shop coverage.

b. Mesh mode which performs similarity-gated expert activation using cosine distance between the mission and MOS-specific embedding.

c. Hybrid mode, which uses Org for deterministic status retrieval and Mesh for cross-functional planning and COA generation.

(3) The system maps agents into a multi-dimensional vector space by embedding each billet's MOS-scoped T&R tasks, METL tasks, and staff guidance into high-dimensional vectors, effectively turning the staff into a semantic manifold where proximity reflects the similarity between the task and their relevant specialties. The number of agents is determined by the selection of the computational pathway. The Organizational mode will

generate and map the appropriate number of agents to mirror the structure of the organization belonging to the commander. The Mesh mode will activate an appropriate distribution of agents as deemed necessary by an Orchestrator. The total number will be directly related to the mission context provided by the commander. When a commander submits a mission or question, ARCnet embeds that input into the same vector space, producing a mission vector that serves as a query. The system computes cosine similarity between the mission vector and every agent vector, identifying which agents lie closest in this conceptual plane and are therefore most relevant to the problem. This similarity is combined with policy constraints, chain-of-command structure, readiness indicators, and workload balancing in a composite score that governs activation. The highest-scoring agents form a "region of activation," enabling ARCnet to perform relevance-weighted parallel reasoning rather than relying solely on fixed staff lanes. This geometric mapping reduces unnecessary LLM calls, improves contextual relevance, and accelerates planning by activating only the most relevant experts. At the same time, "must-include" shops and policy rules ensure that mandatory perspectives such as operations, logistics, or fiscal always appear regardless of similarity.

(4) Activated agents then perform bounded, schema-controlled reasoning over the Mission Context, MOS-scoped doctrine, and the unclassified Single Source of Truth (current readiness, maintenance, funding, TEEP). Specialists also incorporate ML forecasts (maintenance backlogs, availability trajectories, risk windows, and contracted support) derived from lightweight time-series models. Their outputs contain structured evidence citations, confidence scores, and classification markings, forming the base units of ARCnet's explainability and auditability.

(5) ARCnet then compiles these specialist assessments, resolving contradictions and surfacing conflicts (e.g., TEEP impacts or maintenance-resource shortfalls). It then checks leadership imposed constraints and determines whether a COA tournament is required. For planning tasks, the COA Generator produces multiple, rubric-scored options with quantified readiness, funding, and scheduling effects, grounded in doctrine and the SSOT. Once a COA is selected, the Tasking Agent produces a sequenced, Unit-aligned operational plan.

(6) All stages emit checkpoints—structured, human-readable and machine-auditable records that preserve evidence, classification metadata, and HITL approvals. These checkpoints form ARCnet's transparent "decision ledger".

(b) Testing and Validation.

(1) Golden Scenario test performance results in repeatable and correct outcomes.

(2) COA ranking consistency for repeated scenarios with expected outcomes.

(3) Token budget level adjustment and performance impacts.

(4) Red-team/Violation test to determine if ARCnet will violate a constraint or policy restriction.

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(c) Schedule and Milestones.

Pre-Start (Nov-Dec 2025)	Compile unclassified data, finalize scope and metrics.
Month 1 (Jan 2026)	Baseline status dashboard, and first end-to-end flow.
Month 2 (Feb 2026)	Train maintenance/readiness forecasting v1
Month 3 (Mar 2026)	Add Mesh/Hybrid, run scenarios; measure accuracy/timeliness
Month 4 (Apr 2026)	Improve ML forecasts; refine weights; draft publishable paper
Month 5 (May-Jun 2026)	Final demos and submission

CONCLUSION. For 1st MLG, readiness levels depend on balancing resources versus training and operational demands. ARCnet provides deep and auditable insights into the state of the command and its assets. It will speed up cross-functional decision making by bringing the right experts together (virtually). The approach is portable, customizable, controlled by the commander, and built to respect policy and security constraints. Easily adaptable to other commands and organizational structures.


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ARCnet Agent Selection

Mesh and Hybrid Modes

Notation

$\mathbf{v}_m$: mission/request embedding

$\mathbf{v}_i$: agent/billet doctrine centroid

$\cos(\cdot, \cdot)$: cosine similarity

$\text{Chain}(i)$, $\text{Coverage}(i)$, $\text{Avail}(i)$, $\text{Load}(i) \in [0, 1]$

K_{total} , K_g : overall and per-shop caps

τ_{sim} , τ_{avail} : similarity and availability thresholds

$\lambda \geq 0$: diversity (redundancy) penalty

Composite Score

$$S(i) = w_{\text{sim}} \cos(\mathbf{v}_m, \mathbf{v}_i) + w_{\text{chain}} \text{Chain}(i) + w_{\text{policy}} \text{Coverage}(i) + w_{\text{ready}} \text{Avail}(i) - w_{\text{load}} \text{Load}(i). \quad (1)$$

A billet's overall fit for this mission. Start with embedding similarity (how relevant the billet is to the mission text), then add governance factors (chain-of-command fit and policy/coverage). Prefer agents who are available and lightly loaded; current workload subtracts from the score. Weights tune the importance of each factor.

Candidate Pool

$$\mathcal{C} = \{ i \in \mathcal{A} : \cos(\mathbf{v}_m, \mathbf{v}_i) \geq \tau_{\text{sim}}, \text{Avail}(i) \geq \tau_{\text{avail}} \}. \quad (2)$$

Before selecting, screen out agents that are not similar enough to the mission or are insufficiently available. Only relevant and available agents enter the pool.

Mesh Mode

Let $M = \text{MustInclude}$. Select additional agents by maximizing score with a redundancy penalty under caps and coverage:

$$A_{\text{mesh}} = M \cup \underset{A \subseteq \mathcal{C} \setminus M}{\text{argmax}} \left[\sum_{i \in A} S(i) - \lambda \sum_{\substack{i < j \\ i, j \in A}} \cos(\mathbf{v}_i, \mathbf{v}_j) \right] \quad (3)$$

Start with mandated shops. Then add the set of agents that maximizes total score while discouraging redundant picks (penalize adding agents that are too similar to each other). This yields a small, relevant, diverse team.

subject to

$$|A| \leq K_{\text{total}} - |M|, \quad |A_g| \leq K_g - |M_g| \quad (\forall g), \quad \text{required-shop coverage satisfied.} \quad (4)$$

Enforce overall and per-shop caps and ensure required coverage across key shops.

Hybrid Mode

Let A_{org} be the deterministic org-lane set. Use residual capacity and exclude already-selected agents:

$$K' = K_{\text{total}} - |A_{\text{org}}|, \quad \mathcal{C}' = \mathcal{C} \setminus A_{\text{org}}. \quad (5)$$

$$A_{\text{hybrid}} = A_{\text{org}} \cup \underset{A \subseteq \mathcal{C}'}{\text{argmax}} \left[\sum_{i \in A} S(i) - \lambda \sum_{\substack{i < j \\ i, j \in A}} \cos(\mathbf{v}_i, \mathbf{v}_j) \right]. \quad (6)$$

Keep the org-lane agents for status retrieval, then use remaining capacity to add the best Mesh-selected specialists. Do not re-select agents already in the org set.

subject to

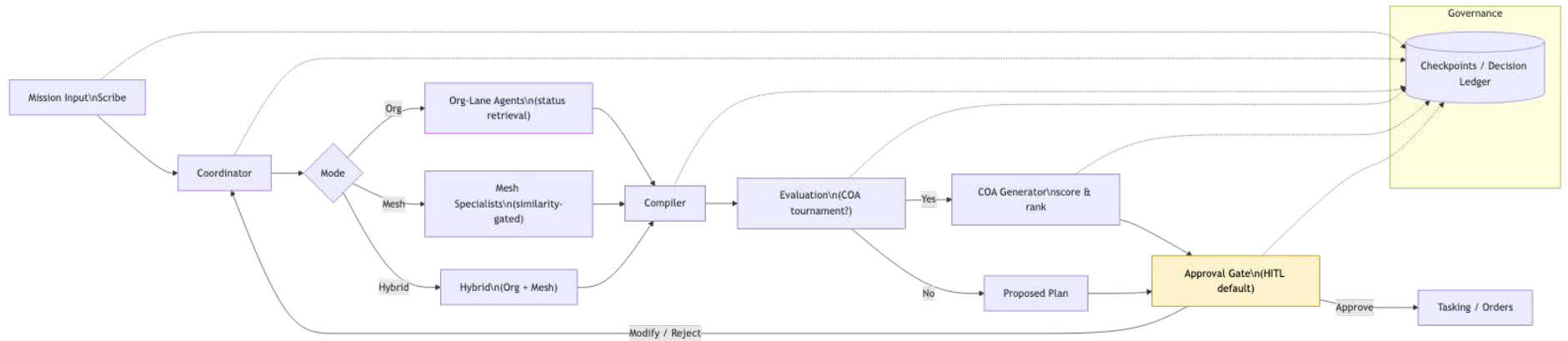
$$|A| \leq K', \quad |A_g| \leq K_g - |(A_{\text{org}})_g| \quad (\forall g), \quad \text{required-shop coverage satisfied.} \quad (7)$$

Apply the same caps and coverage rules, but with reduced capacity after accounting for the org-lane picks.

Slide-Simple

$$A_{\text{active}} = \text{MustInclude} \cup \text{TopK}_{\text{constrained}}(S(i)). \quad (8)$$

Commander's Input



COA Tournament

